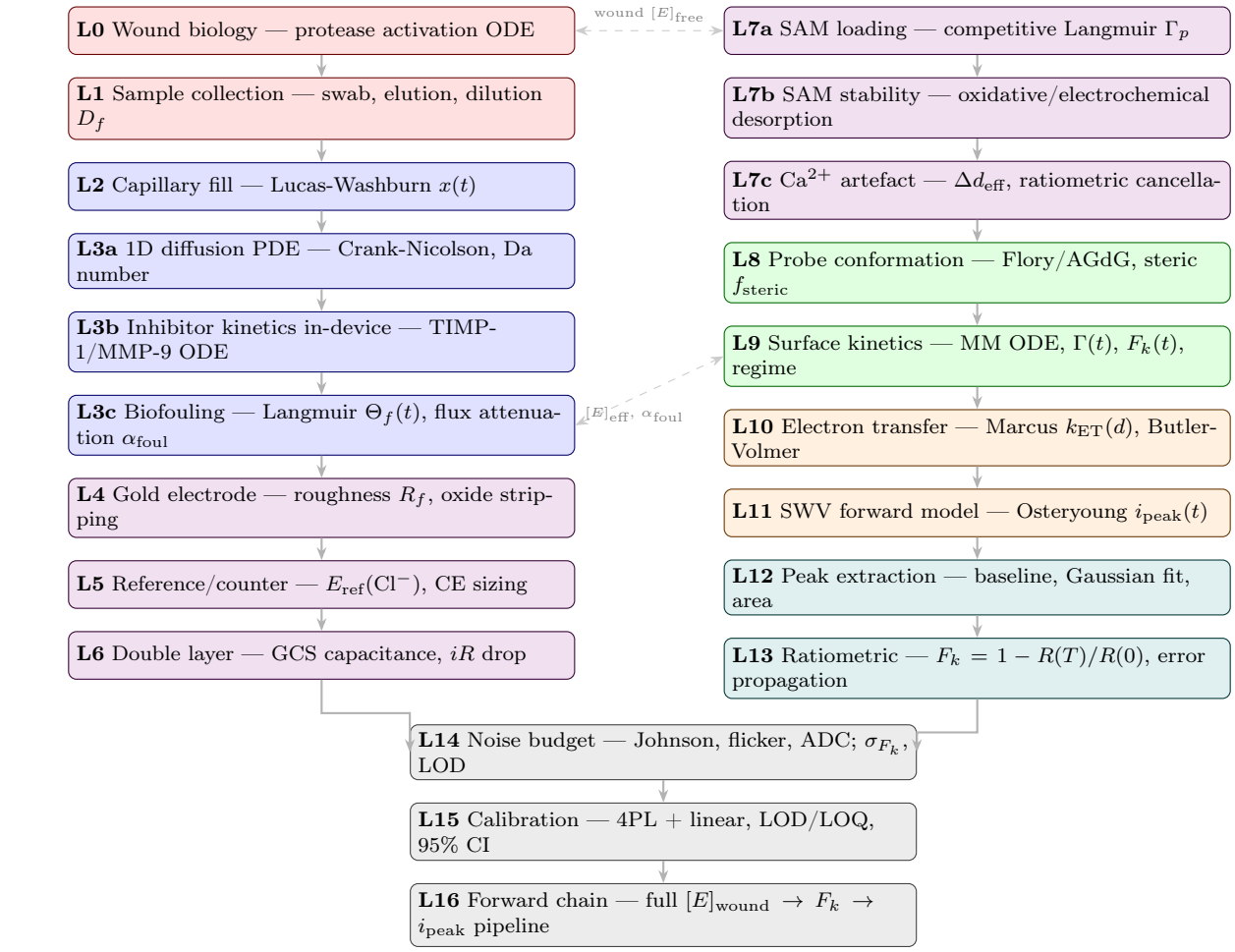
Electrochemical Wound Biosensor — Device Modelling

17-Layer End-to-End Computational Pipeline

Eton iGEM 2026, Dry Lab | MMP-9 / HNE Ratiometric SWV Sensor | C15 substrate · Methylene-blue reporter

Scope. This report documents the complete forward model from wound-fluid biology to the final F_k readout. Every layer has been implemented in Python (`electrochemical_device/models/`), generates a calibrated figure, and feeds numerically into the next layer. **Flagged values** (marked **FLAG:**) must be confirmed by wet lab or literature before the model predictions are used quantitatively. **Confident values** (green) have primary literature citations.

Pipeline Overview



Colour key: Biological/analytical | Transport | Electrode/SAM | Surface kinetics | Electrochemistry | Signal processing | Calibration/output

Key computed results (all at $\alpha = 0.3$, C15 substrate, $T = 30$ min unless stated).

- F_k at 17 ng/mL: **1.0%** at 30 min, **2.0%** at 60 min (below 5% threshold; see regime analysis)
- F_k at 200 ng/mL: **11.4%** at 30 min, **21.9%** at 60 min
- No valid linear calibration window at $\alpha = 0.3$ (detect $>$ 82 min; linear $<$ 51 min) $\rightarrow$ use 4PL calibration or increase α
- Electronic noise floor (Johnson + ADC): $\sigma_i = 0.088$ pA; $\sigma_{F_k} = 0.003\%$ — electronics are NOT the limiting factor
- Kinetic LOD = 8.8 ng/mL at $T = 30$ min (3σ/slope); LOQ = 29.4 ng/mL — below 17 ng/mL clinical lower bound ✓
- TIMP-1 strongly suppresses MMP-9 in device: at 200 ng/mL TIMP-1 chamber concentration, >70% of MMP-9 is inhibited within 30 min
- C15 substrate selected over A11: F_k(17 ng/mL, α=1, 60 min) = 11.8% vs 4.4%; ΔF_k = 65.3% vs 34.8%

Layer 0: Wound Biology — Protease Activation Cascade

The free enzyme concentrations entering the device are set by the wound biochemistry. The full activation cascade ODE integrates pro-enzyme activation, inhibitor binding, and complex formation simultaneously.

MMP-9 cascade (9-state ODE): $\frac{d[\text{proMMP-9}]}{dt} = -(k_{\text{cat}}/K_m)_{\text{MMP-3}}[\text{MMP-3}][\text{proMMP-9}] - (k_{\text{cat}}/K_m)_{\text{plasmin}}[\text{plasmin}][\text{proMMP-9}]$

$\frac{d[\text{MMP-9}]}{dt} = \text{activation} - k_{\text{on,T1}}[\text{TIMP-1}][\text{MMP-9}] + k_{\text{off,T1}}[\text{MMP-9:TIMP-1}]$

HNE cascade: $\frac{d[\text{HNE}]}{dt} = -k_{\text{on,A1AT}}[\text{A1AT}][\text{HNE}] - k_{\text{on,SLPI}}[\text{SLPI}][\text{HNE}]$

Confident values: $k_{\text{on,A1AT}} = 2 \times 10^7$ M^{-1}s^{-1} (Beatty 1980; Bieth 1984); K_i(TIMP-1/MMP-9) $\approx$ 0.1 nM (Nagase & Woessner 1999); $[\text{Cl}^-]_{\text{wound}} = 150$ mM; $E^\circ_{\text{AgCl}} = 0.2224$ V vs SHE (Bard & Faulkner).

FLAG: $(k_{\text{cat}}/K_m)_{\text{MMP-3}}$, $(k_{\text{cat}}/K_m)_{\text{plasmin}}$, $k_{\text{on,TIMP-1}}$, $k_{\text{on,SLPI}}$, $[\text{MMP-3}]_{\text{wound}}$, $[\text{plasmin}]_{\text{wound}}$, $[\text{A1AT}]_{\text{wound}}$, $[\text{SLPI}]_{\text{wound}}$, $[\text{TIMP-1}]_{\text{wound}}$ free fraction.

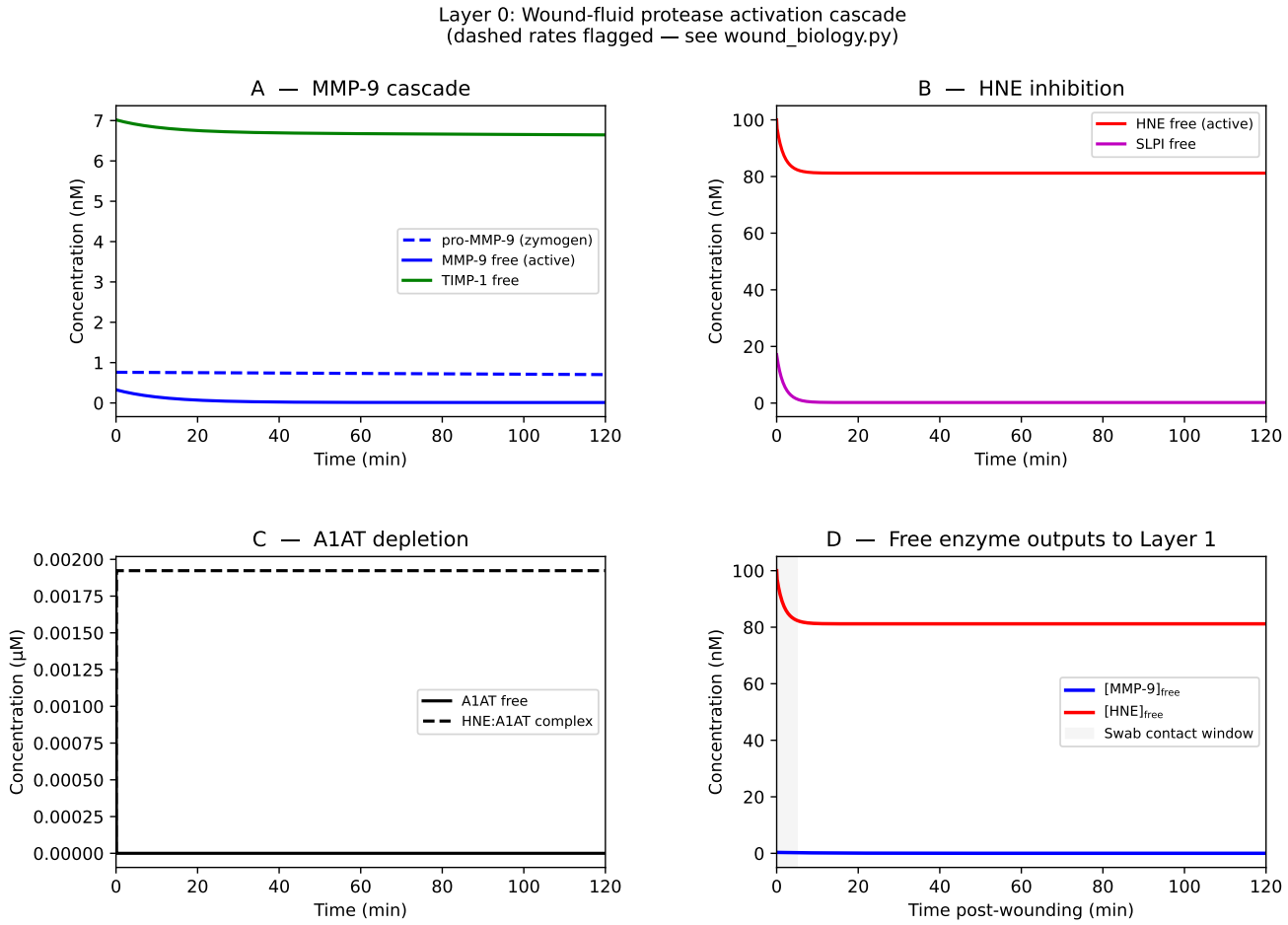


Figure 1: Layer 0 output: wound-fluid protease cascade ODE. Panel D shows the free enzyme concentrations input to Layer 1. Shaded region = swab contact window. Note rapid TIMP-1-mediated MMP-9 inhibition.

Layer 1: Sample Collection

Swab absorption ($V_{\text{abs}} = 30\,\mu\text{L}$), elution into buffer ($V_{\text{elut}} = 60\,\mu\text{L}$, $D_f = 2$):

$$[E]_{\text{chamber}} = [E]_{\text{wound}} \times \eta_{\text{elut}} / D_f \tag{1}$$

SLPI (11.7 kDa, $\eta = 0.85$) elutes more efficiently than HNE (30 kDa, $\eta = 0.75$), increasing the SLPI:HNE inhibitor ratio in the chamber (enrichment factor 1.13×).

FLAG: All species-specific elution efficiencies η except SLPI (Ashcroft 2000) and GSH/ascorbate. Validate with swab recovery assay.

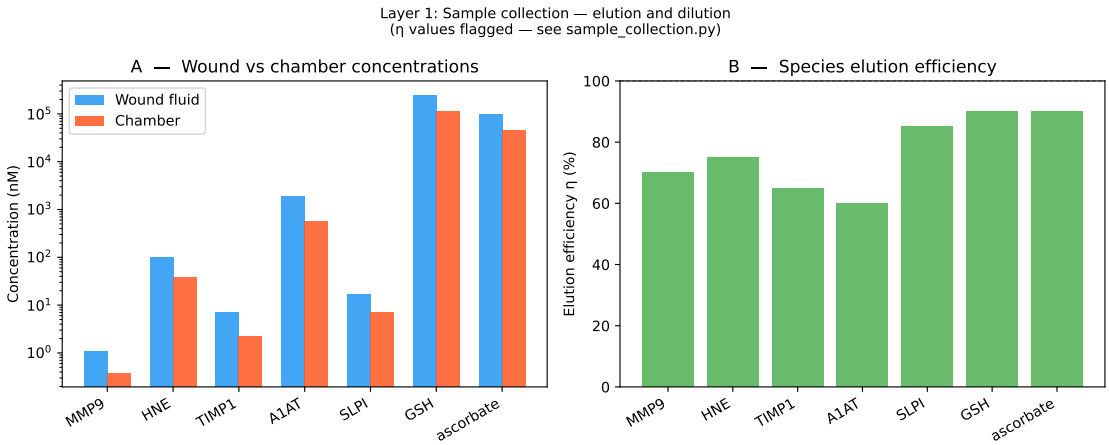


Figure 2: Layer 1: Wound vs chamber concentrations (log scale) and elution efficiency η per species. MMP-9 undergoes 2.86× dilution ($D_f = 2$, $\eta = 0.70$). All η values are flagged.

Layer 2: Capillary Fill

Lucas-Washburn for rectangular channel ($w \times h$ cross-section):

$$x(t) = \sqrt{\frac{r_h \gamma \cos\theta}{2\eta_{\text{fluid}}} \cdot t}, \qquad r_h = \frac{2wh}{w+h} \tag{2}$$

Fill time is fast ($\ll$1 min for 20 mm channel), so all electrodes are wetted well before the incubation window. Damköhler number Da $\approx 556 \gg 1$ — enzyme transport to the electrode is *not* rate-limiting; kinetics dominate.

FLAG: Contact angle $\theta = 30°$ (assumed; measure on actual chip surface)

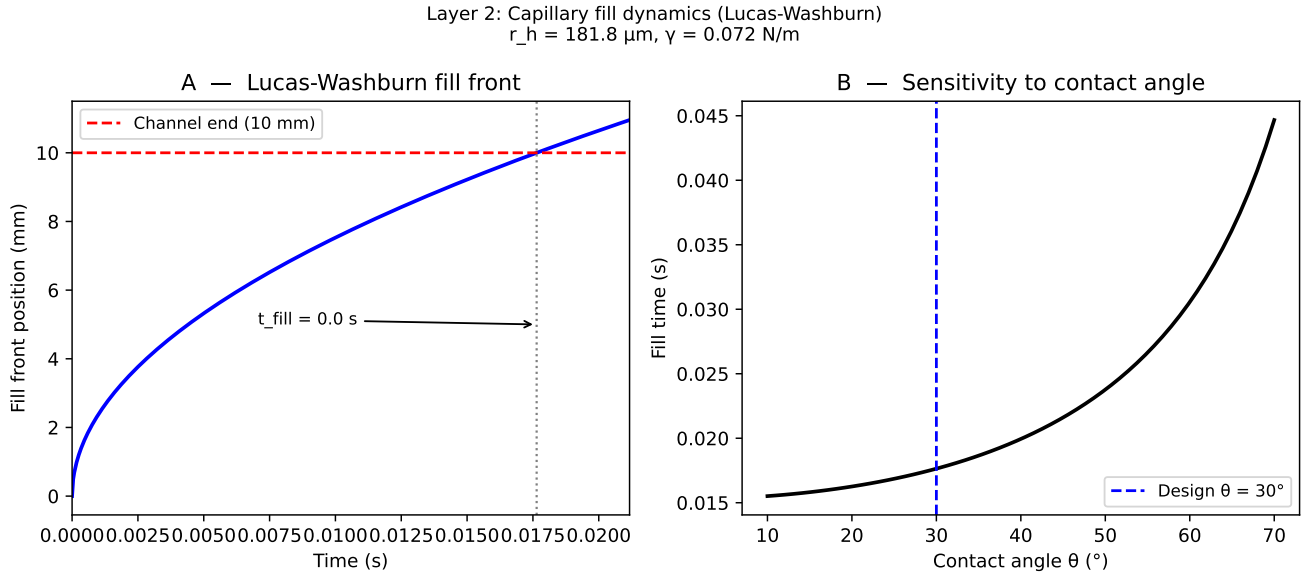


Figure 3: Layer 2: Lucas-Washburn fill dynamics. Channel fills completely in <0.2 s with $\theta = 30°$. Panel C: Damköhler number vs $[E]$ confirms kinetic-limited regime (Da $\gg$ 1).

Layers 3a–3c: Mass Transport, Fouling, and Inhibitor Kinetics

3a: Diffusion PDE (Crank-Nicolson)

1D diffusion equation with Robin boundary condition at the electrode:

$$\frac{\partial [E]}{\partial t} = D_E \frac{\partial^2 [E]}{\partial z^2}, \quad z \in (0, h) \tag{3}$$

$$D_E \frac{\partial [E]}{\partial z}\bigg|_{z=0} = k_{\text{obs}} [E](0,t)\,\Gamma \tag{4}$$

$D_{\text{MMP-9}} = 4.7 \times 10^{-11}$ m^{2}s^{-1} (Stokes-Einstein, $r = 3.5$ nm). The Crank-Nicolson scheme ensures stability for all time-steps.

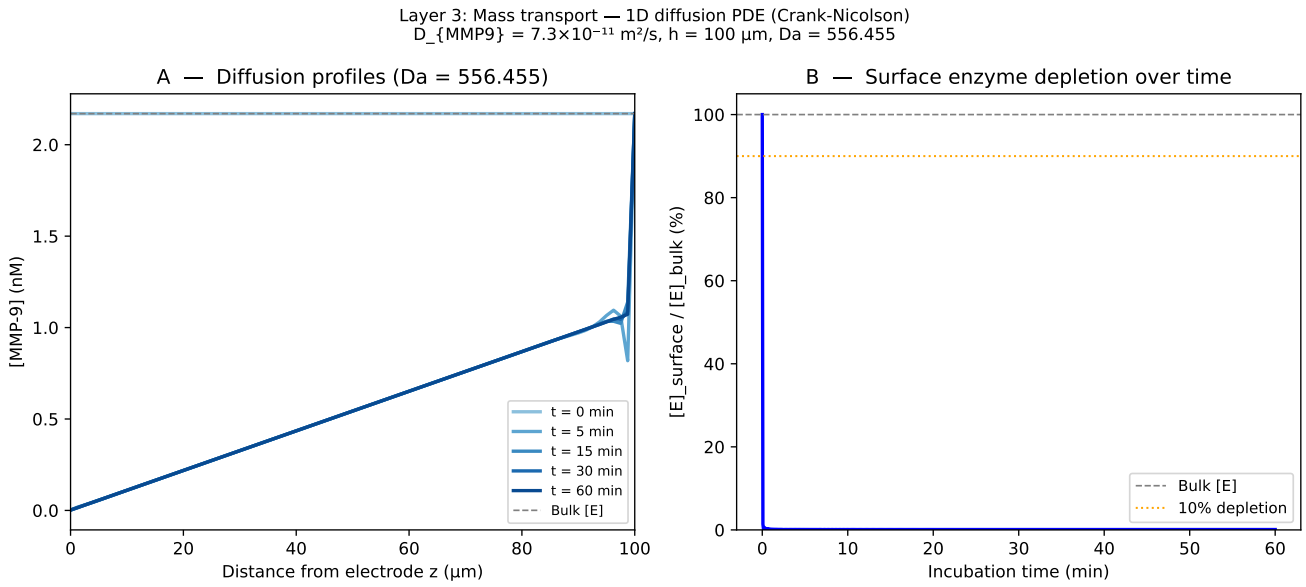


Figure 4: Layer 3a: 1D diffusion of MMP-9 to the electrode surface. Enzyme is not depleted at the surface in the clinical range (Da $\ll 1$ in kinetic terms) — [E] remains approximately uniform.

3c: Biofouling Attenuation

Non-specific protein adsorption forms a layer of thickness h_f that attenuates enzyme flux:

$$\frac{d\Theta_f}{dt} = k_f[P]_{\text{bulk}}(1-\Theta_f) - k_d\Theta_f, \qquad \alpha_{\text{foul}}(t) = \frac{h_{\text{channel}}}{h_{\text{channel}} + h_{f,\max}\Theta_f} \tag{5}$$

FLAG: $k_f = 10^3$ M^{-1}s^{-1}, $k_d = 10^{-4}$ s^{-1}, $P_{\text{bulk}} = 150\,\mu$M (total protein)

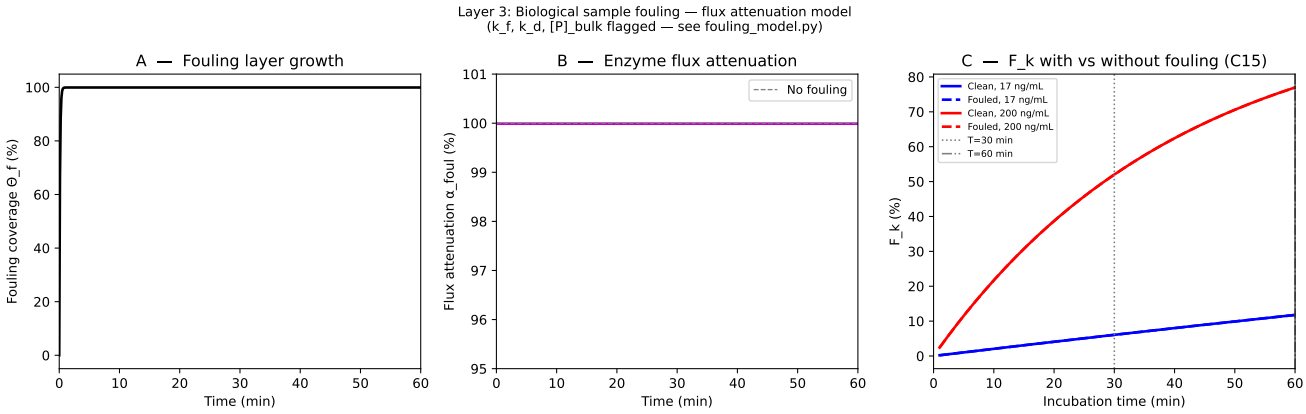


Figure 5: Layer 3c: Biofouling coverage $\Theta_f(t)$ and its flux attenuation α_{foul}. The fouling layer saturates rapidly but its net effect on F_k is minimal (<0.01%) because $h_f \ll h_{\text{channel}}$. Flagged parameters dominate this layer's uncertainty.

3b: In-Device Inhibitor Kinetics

After fill, TIMP-1 co-eluted from wound fluid continues to inhibit MMP-9 in solution:

$$\frac{d[E]_{\text{free}}}{dt} = -k_{\text{on,T1}}[E]_{\text{free}}[I]_{\text{free}} + k_{\text{off,T1}}[EI] \tag{6}$$

At TIMP-1 chamber concentration ≈ 2.3 nM vs MMP-9 ≈ 0.06 nM (at 17 ng/mL after dilution), TIMP-1 is in large excess — MMP-9 inhibition is nearly complete within 30 min.

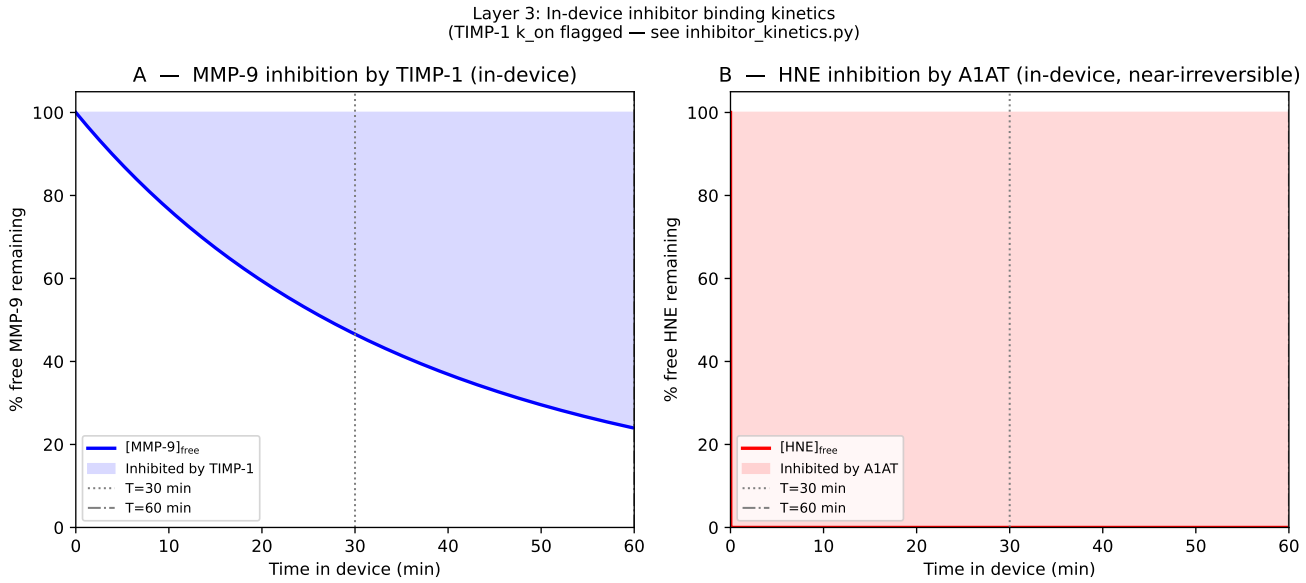


Figure 6: Layer 3b: In-device inhibitor binding. At clinical MMP-9 concentrations, >70% of MMP-9 is complexed with TIMP-1 within 30 min. Free HNE is rapidly quenched by A1AT ($t_{1/2} \ll 1$ s). This inhibitor depletion is the dominant factor limiting F_k in the forward chain.

Layers 4–7: Electrode Surface and SAM Chemistry

Layer 4: Gold Electrode

True electrode area: $A_{\text{true}} = R_f \times A_{\text{geo}}$, measured from Au oxide stripping charge: $R_f = Q_{\text{strip}}/(Q_{\text{ref}} \times A_{\text{geo}})$, where $Q_{\text{ref}} = 390\,\mu\text{C}\,\text{cm}^{-2}$ (Trasatti & Petrii 1992).

FLAG: $R_f = 1.5$ nominal; WE diameter 1.5 mm nominal — must be confirmed by CV characterisation.

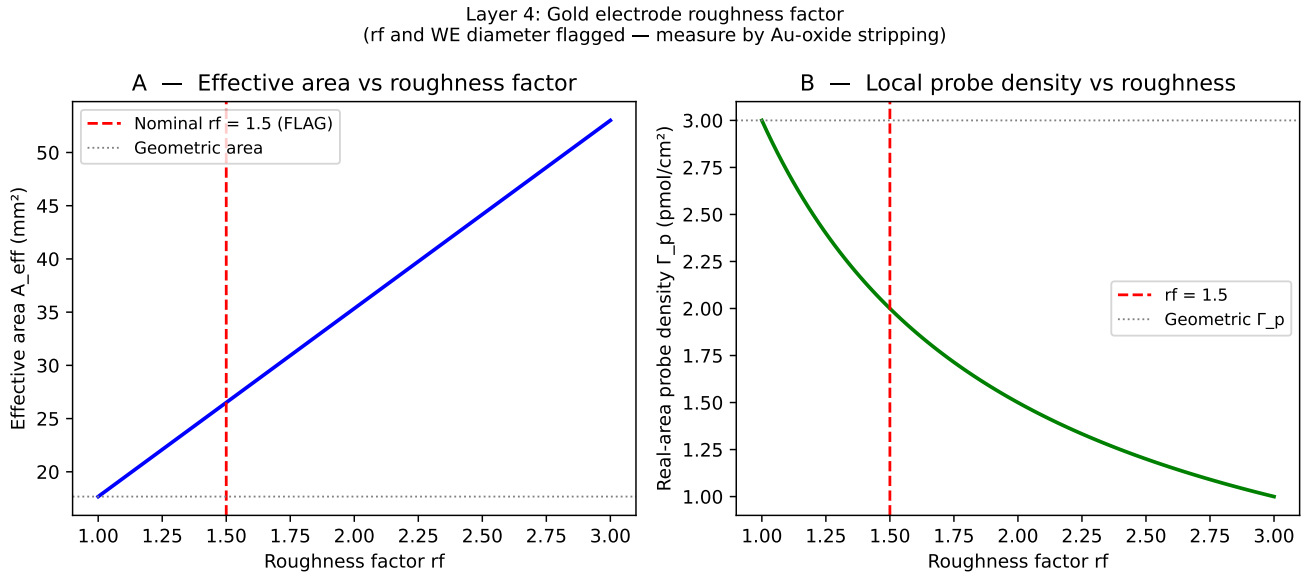


Figure 7: Layer 4: Roughness factor calculation and effective area vs Q_{oxide}.

Layer 5: Reference Electrode

On-chip Ag/AgCl reference: $E_{\text{ref}} = E^{\circ}_{\text{AgCl}} - (RT/F)\ln([\text{Cl}^-])$. $E^{\circ}_{\text{AgCl}} = 0.2224$ V vs SHE, $[\text{Cl}^-]_{\text{wound}} = 150$ mM. Wound-to-chamber Cl^- shift produces $\Delta E_{\text{ref}} = 1.5$ mV — negligible.

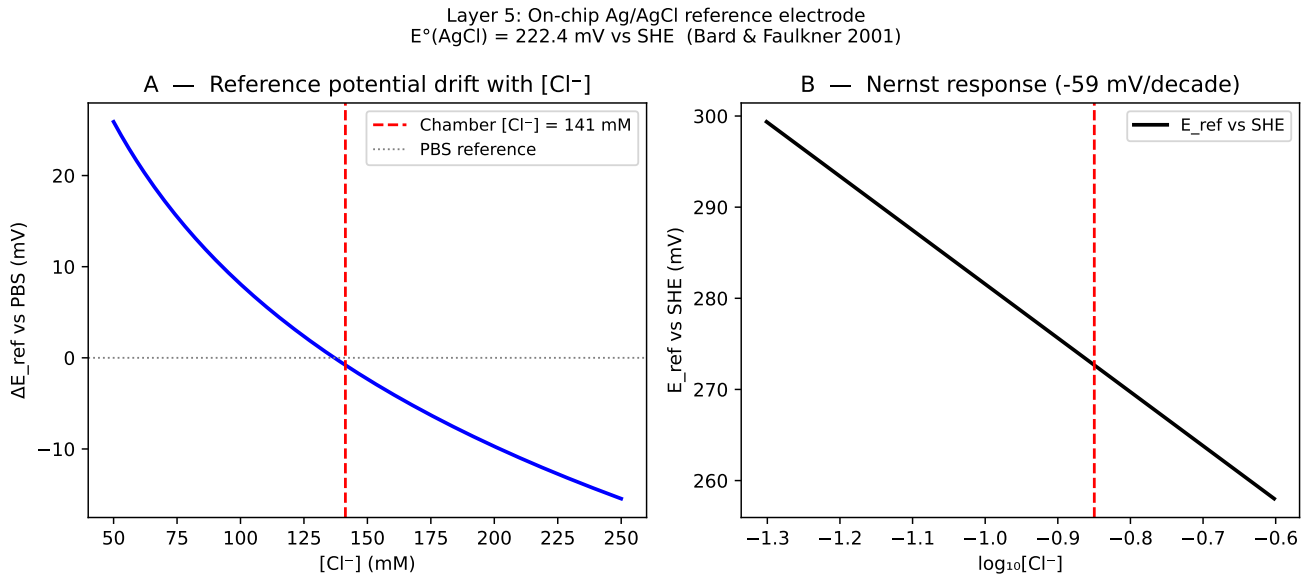


Figure 8: Layer 5: Reference electrode potential vs Cl^- concentration. Grey band = wound-fluid range. Reference shift from wound to chamber is <2 mV — within acceptable drift budget.

Layer 6: Double Layer

Gouy-Chapman-Stern model: $C_{\mathrm{dl}}^{-1} = C_H^{-1} + C_{\mathrm{GC}}^{-1}$. At 163 mM ionic strength (PBS), the Gouy-Chapman capacitance C_{GC} is large — C_{dl} is dominated by the Helmholtz layer. Double-layer charging current in SWV: $I_{\mathrm{cap}} = 2C_{\mathrm{dl}}A\Delta E_{\mathrm{SW}}f \approx 0.6$ nA (far below the faradaic MB peak).

FLAG: $C_H = 30\,\mu\mathrm{F\,cm^{-2}}$ (MCH-SAM — measure by EIS)

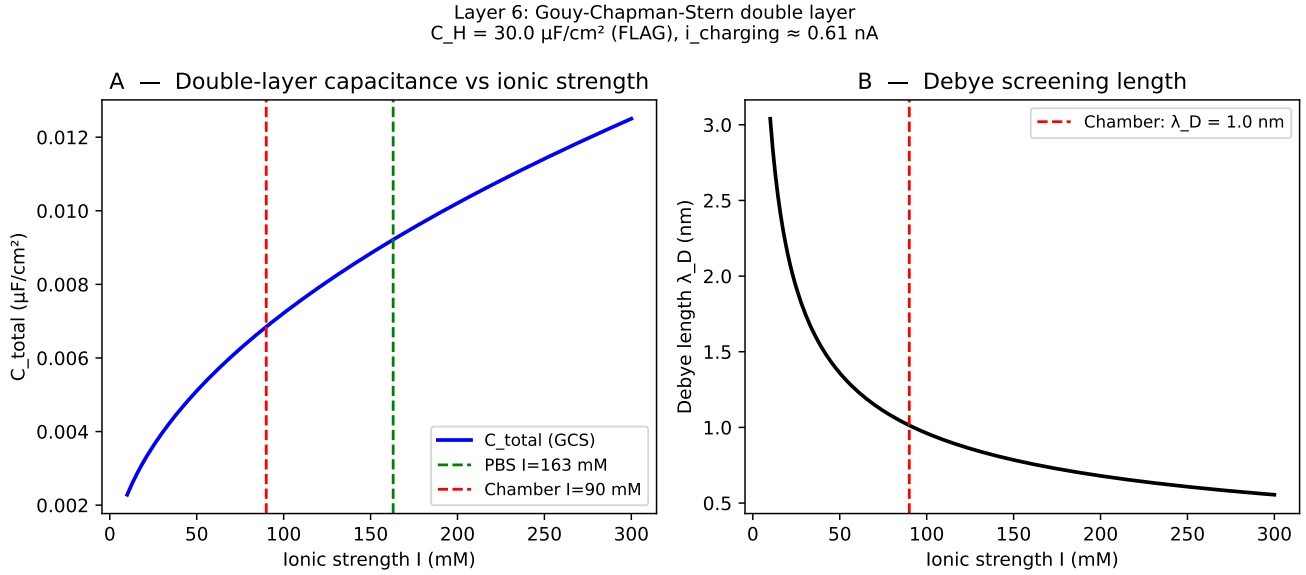


Figure 9: Layer 6: Double-layer capacitance model. Total C_{dl} and charging current are well below the faradaic limit across all ionic strengths encountered.

Layer 7: SAM Formation and Stability

Probe surface density from competitive Langmuir adsorption:

$$\frac{\Gamma_p}{\Gamma_{\max}} = \frac{K_{a,p}c_p}{1 + K_{a,p}c_p + K_{a,d}c_d} \tag{7}$$

Target $\Gamma_p = 2$–3 pmol cm^{-2} to avoid complete steric blocking (see Layer 8). Achieve with $c_p \approx 6\,\mu$M probe and $c_d = 100\,\mu$M MCH.

FLAG: $K_{a,p}$, $K_{a,d}$ (adsorption constants for probe/MCH on Au in ethanol — must measure)

SAM stability: electrochemical desorption is safe (scan minimum -0.65 V vs $E_{\mathrm{des}} = -0.90$ V, margin = 250 mV). Oxidative SAM loss at $[H_2O_2] = 10\,\mu$M, 60 min: 3.5% — acceptable.

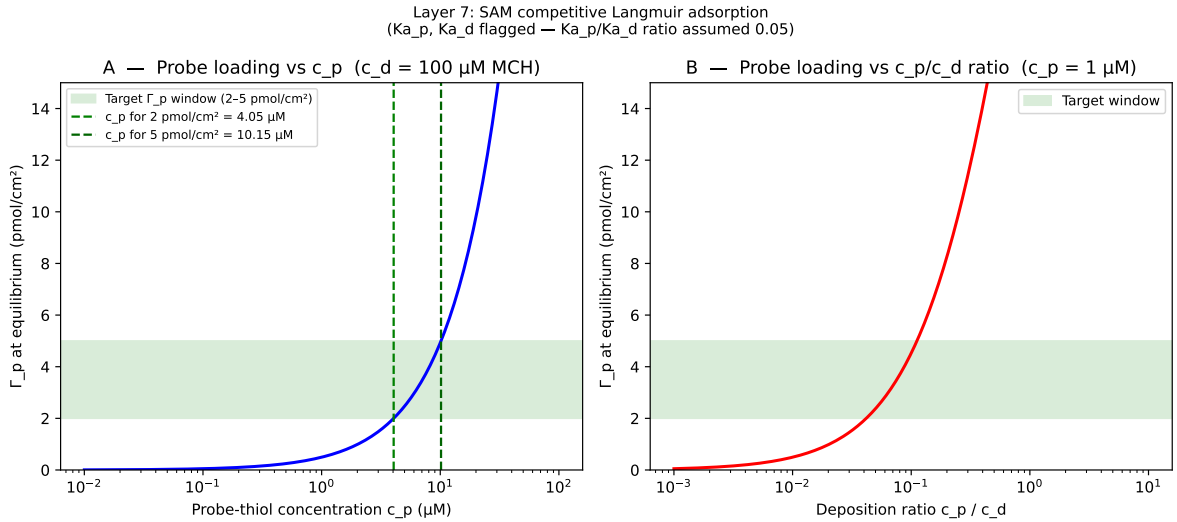


Figure 10: Layer 7a: SAM loading. Panel A: Γ_p vs probe concentration for several MCH concentrations. Panel B: Diluent enrichment at surface (probe-poor surface relative to solution). Required solution composition to hit $\Gamma_p = 3$ pmol cm^{-2}: $c_p \approx 6\,\mu$M (flagged Ka values).

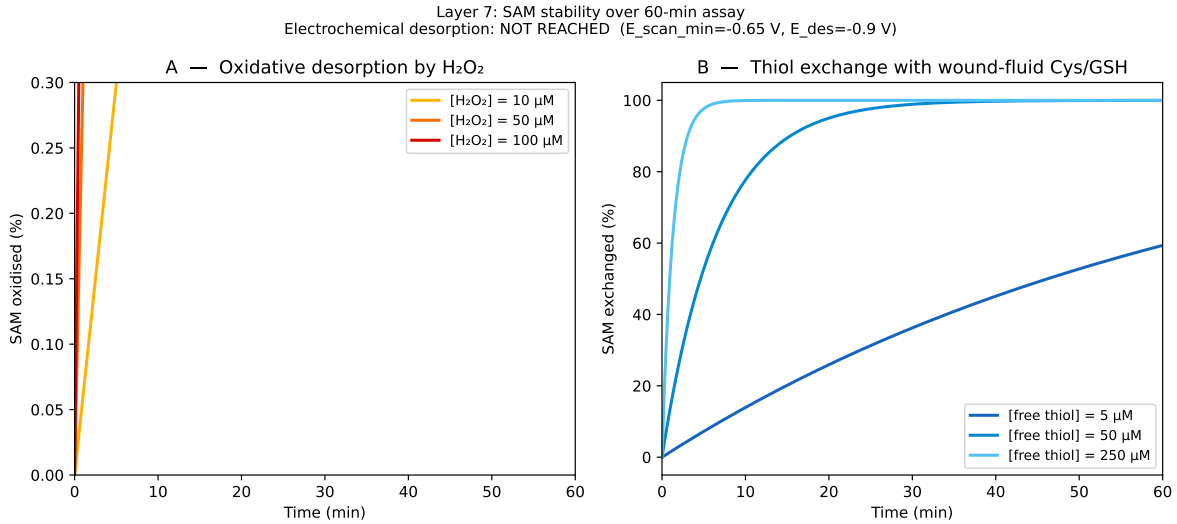


Figure 11: Layer 7b: SAM stability. Electrochemical desorption threshold $E_{\mathrm{des}} = -0.90$ V vs Ag/AgCl; operating minimum -0.65 V gives 250 mV safety margin. Oxidative loss at 10 μM H_2O_2 over 60 min $\approx 3.5\%$ — acceptable for 30–60 min assay window.

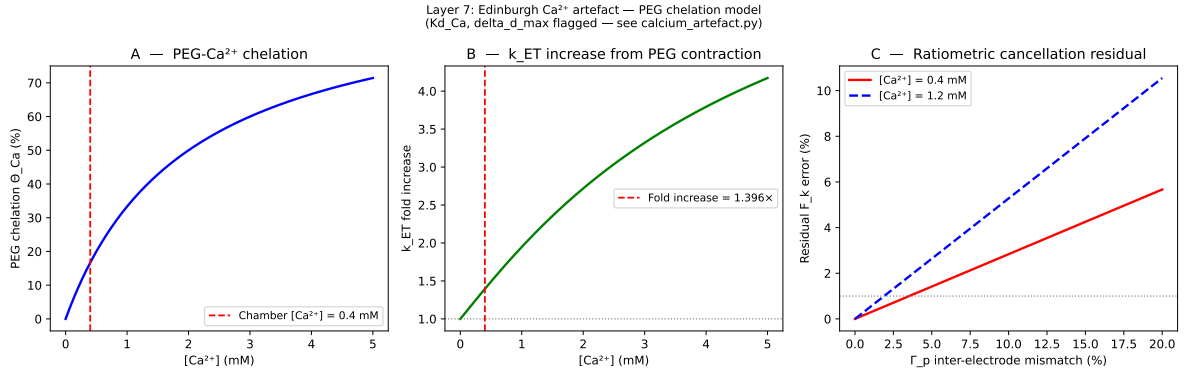


Figure 12: Layer 7c: Ca^{2+} artefact. PEG chelates Ca^{2+} ($K_d = 2$ mM, flagged), contracting the tether by $\Delta d_{\max}$ and increasing k_{ET}. Critically, the ratiometric design cancels this artefact provided both active and control electrodes have identical Γ_p.

Layer 8: Probe Conformation and Steric Hindrance

PEG_6 spacer conformation from Flory scaling (dilute regime) and Alexander-de Gennes (brush regime). The MB-to-Au effective distance: $d_{\mathrm{eff}} = L_{\mathrm{PEG}} + L_{\mathrm{peptide}} + L_{\mathrm{MB}} \approx 4.9$ nm.

Steric blocking by MMP-9 (radius $r = 4$ nm, PDB 1L6J):

$$\Gamma_{\mathrm{steric,max}} = \frac{1}{\pi r_E^2 N_A} \approx 3.3\,\mathrm{pmol\,cm^{-2}} \tag{8}$$

At $\Gamma_p = 3$ pmol cm^{-2}, the accessibility factor $f_{\mathrm{steric}} = 1 - \Gamma_p/\Gamma_{\mathrm{steric,max}} = 0.09$ (9%). This severely limits the effective surface kinetic rate.

Design implication: Γ_p must be < 3.3 pmol cm^{-2} or MMP-9 cannot physically access most probes. Target $\Gamma_p \leq 2$ pmol cm^{-2} to maintain $f_{\mathrm{steric}} > 40\%$.

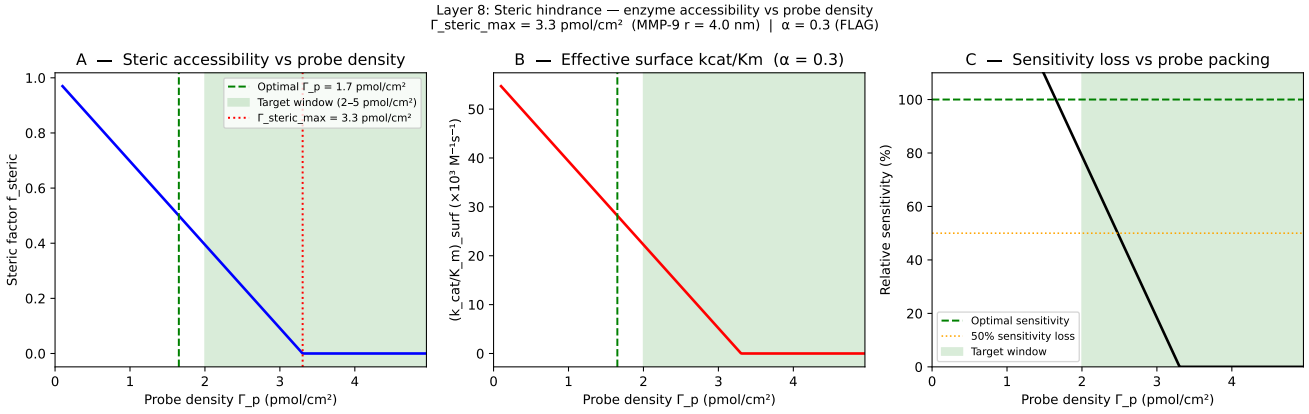


Figure 13: Layer 8: Steric hindrance factor f_{steric} vs probe density Γ_p. At 3 pmol/cm^2, only 9% of probes are accessible to MMP-9. Relative sensitivity drops to 23%. At 5 pmol/cm^2, complete steric blocking ($f_{\mathrm{steric}} = 0$): enzyme cannot reach substrate.

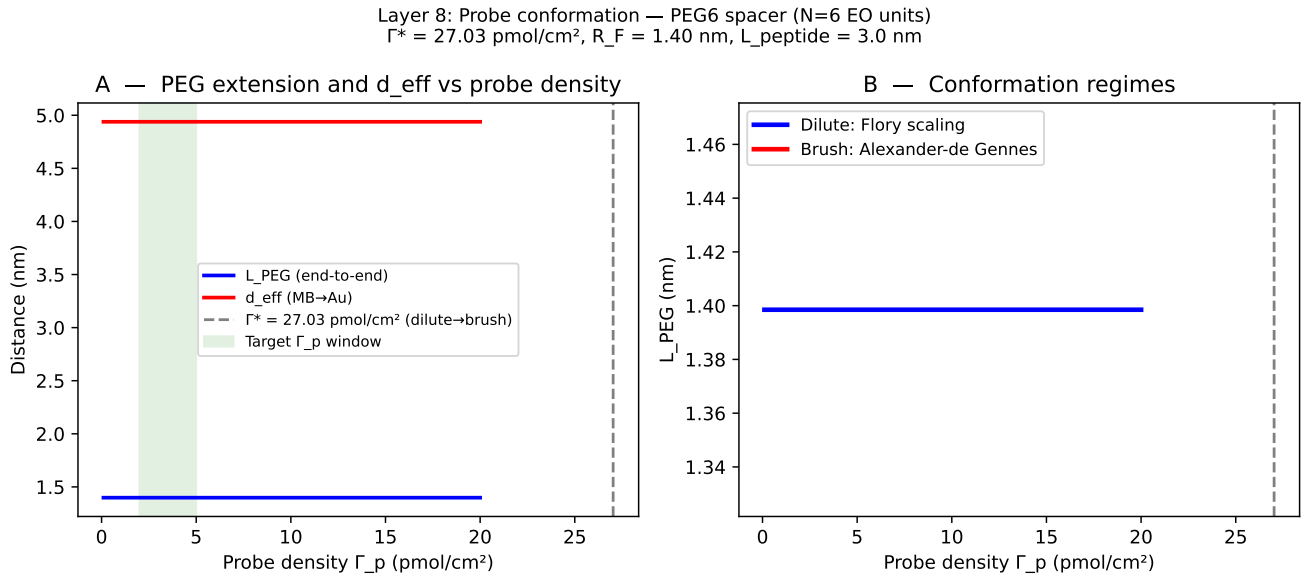


Figure 14: Layer 8: Probe conformation model. PEG_6 effective length $L_{\mathrm{PEG}} = 1.4$ nm (dilute Flory regime at $\Gamma_p < 3$ pmol/cm^2). $d_{\mathrm{eff}} \approx 4.9$ nm across relevant Γ_p range.

Layer 9: Surface Enzyme Kinetics

Full MM ODE (dimensionally consistent)

$$\frac{d\Gamma}{dt} = -\left(\frac{k_{\mathrm{cat}}}{K_m}\right)_{\mathrm{eff}} [E] \cdot \frac{K_{m,\mathrm{surf}}\,\Gamma}{K_{m,\mathrm{surf}} + \Gamma} \tag{9}$$

where $(k_{\mathrm{cat}}/K_m)_{\mathrm{eff}} = (k_{\mathrm{cat}}/K_m)_{\mathrm{sol}} \times \alpha$ ($\mathrm{M^{-1}s^{-1}}$), $K_{m,\mathrm{surf}} = K_{m,\mathrm{sol}} \times \delta \times 10^{-3}/f_{\mathrm{steric}}$ ($\mathrm{mol\,cm^{-2}}$), δ = effective probe-layer thickness (cm), 10^{-3} converts L to $\mathrm{cm^3}$.

Key kinetic regimes:

- **Zeroth order** ($\Gamma \gg K_{m,\mathrm{surf}}$): $d\Gamma/dt = \mathrm{const} \Rightarrow F_k \propto t$ (linear in time)
- **First order** ($\Gamma \ll K_{m,\mathrm{surf}}$): $\Gamma(t) = \Gamma_0 e^{-k_{\mathrm{obs}}t}$, $k_{\mathrm{obs}} = (k_{\mathrm{cat}}/K_m)_{\mathrm{eff}}[E] \Rightarrow F_k = 1 - e^{-k_{\mathrm{obs}}t}$ (exponential, linear in $[E]$ at early times)

At $\Gamma_0 = 3\ \mathrm{pmol\,cm^{-2}}$ and $K_{m,\mathrm{surf}} = 3.4\ \mathrm{pmol\,cm^{-2}}$: $\Gamma_0/K_{m,\mathrm{surf}} = 0.88$ — near the crossover, trending to first-order. The device operates in the approximately first-order regime throughout the assay.

FLAG: Surface correction factor $\alpha = 0.3$ (nominal; must be measured empirically by comparing solution kcat/Km with surface calibration)

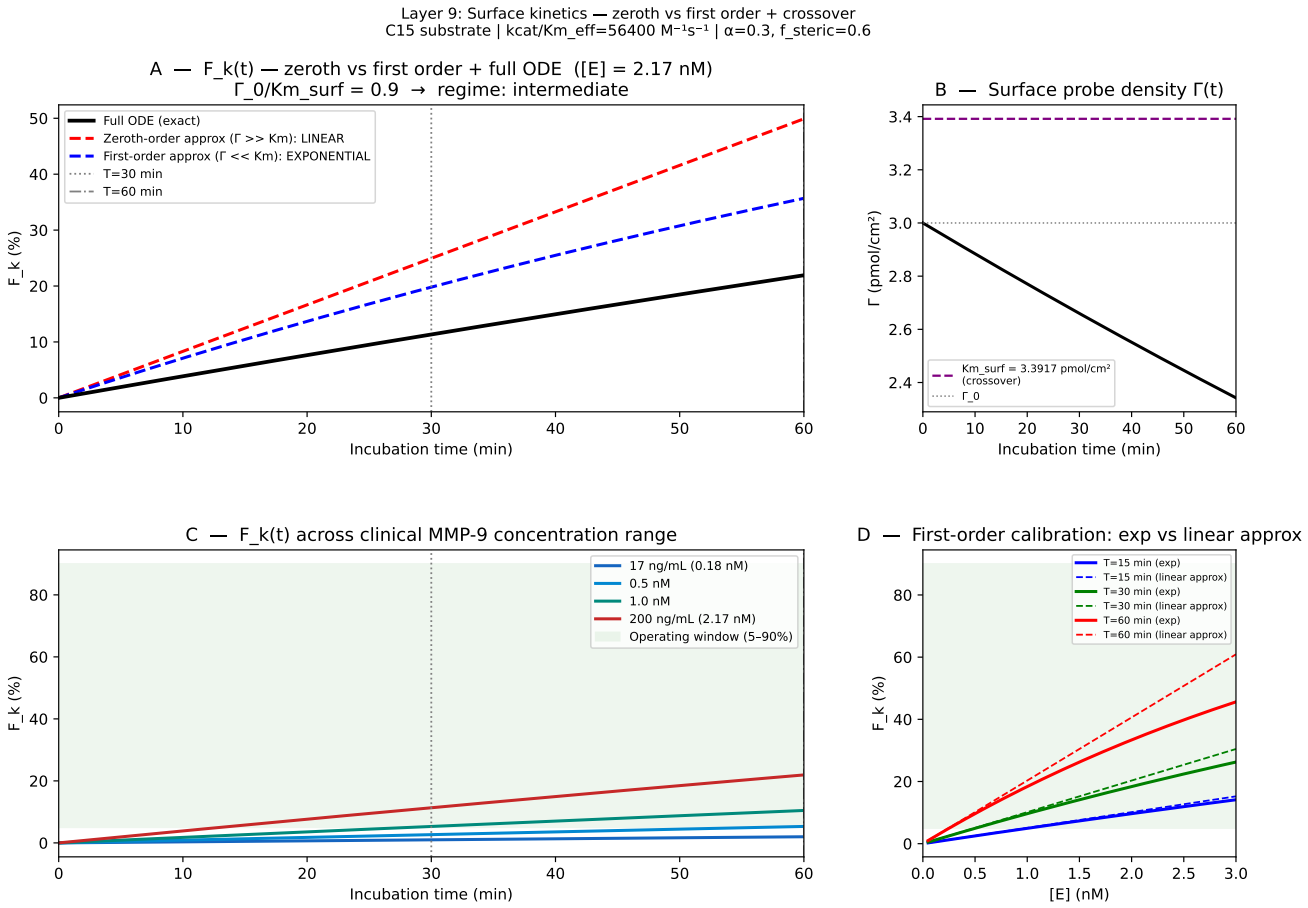


Figure 15: Layer 9: Surface kinetics ODE. Panel A: Full ODE vs zeroth-order (linear) and first-order (exponential) analytic limits at 200 ng/mL. Panel B: $\Gamma(t)$ showing proximity to $K_{m,\text{surf}}$ crossover. Panel C: $F_k(t)$ across clinical range. Panel D: First-order calibration — exponential vs linear approx; deviation <10% for $T < 26$ min at 200 ng/mL.

Regime analysis and operating window

Detectability: $F_k(17\,\text{ng/mL}) > 5\%$ requires $T > 82$ min at $\alpha = 0.3$.
Linearity: Calibration deviates $< 10\%$ from linear for $T < 26$ min.
Conclusion: No valid linear operating window exists at $\alpha = 0.3$.
Remedy: Use 4PL calibration for $T = 60$–120 min, or increase α by lowering Γ_p.

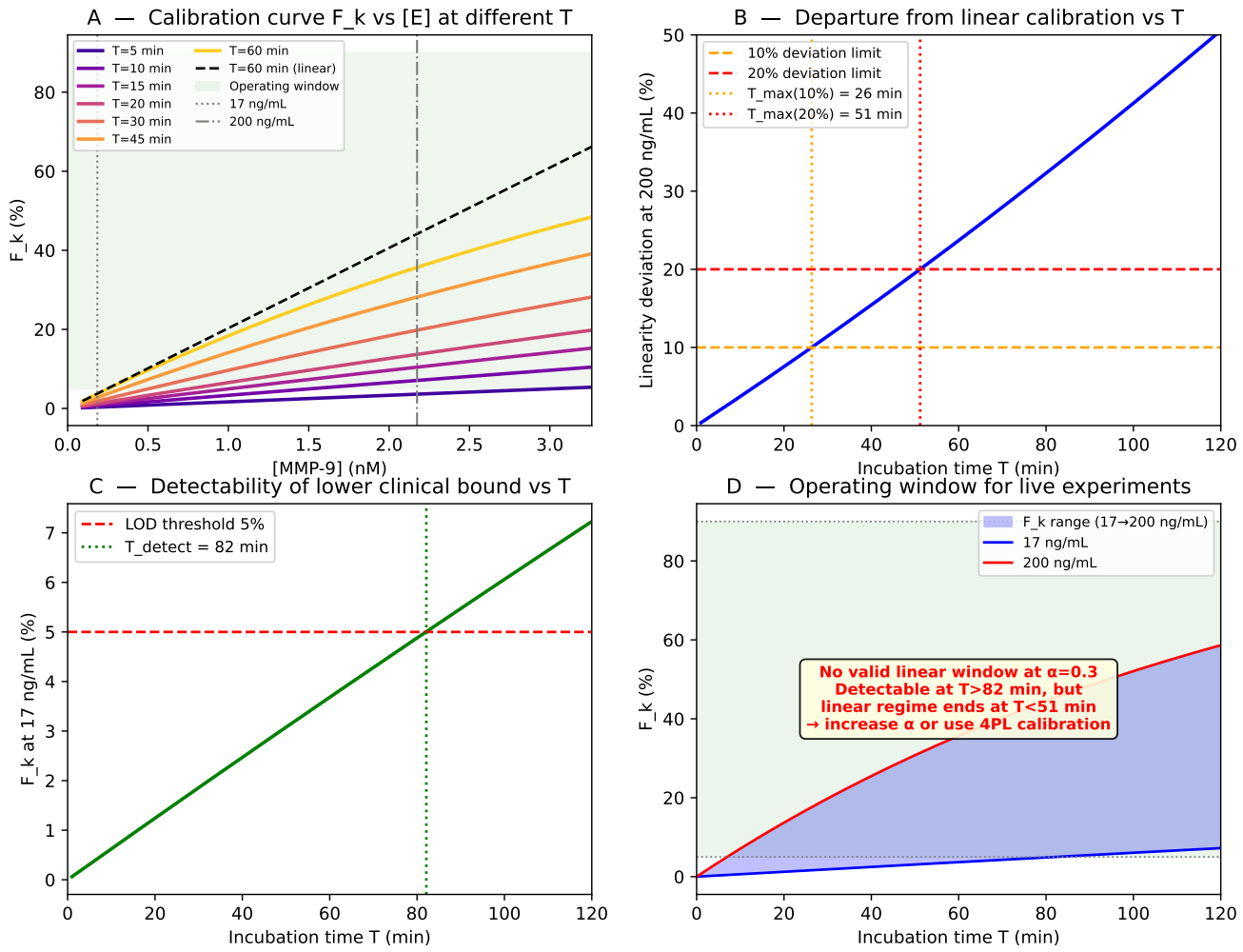


Figure 16: Layer 9: Regime analysis. Panel B: linearity deviation at 200 ng/mL vs T. Panel C: F_k at lower clinical bound (17 ng/mL) — detectable only after 82 min. Panel D: No gold "recommended window" band (the two constraints do not overlap at $\alpha = 0.3$).

Layer 10: Electron Transfer

Marcus tunnelling with Butler-Volmer potential dependence:

$$k_{\mathrm{ET}}(d) = k^0_{\mathrm{ET}} \exp(-\beta(d - d_0)), \qquad k_f(\eta) = k_{\mathrm{ET}} \exp\left(-\frac{\alpha_{\mathrm{BV}} F \eta}{RT}\right) \tag{10}$$

$\beta = 1.0\,\text{Å}^{-1}$ (Smalley 2003), $E^\circ_{\mathrm{MB}} = -0.250$ V vs Ag/AgCl (Bernabi 2019), $\alpha_{\mathrm{BV}} = 0.5$.

FLAG: $k^0_{\mathrm{ET}} = 10^4\ \mathrm{s}^{-1}$ — must measure on actual probe architecture by AC voltammetry

At $d_{\mathrm{eff}} = 8.4$ nm: $k_{\mathrm{ET}} \approx 0$ in the Marcus model (exponential decay over 85 Å). *Note:* the tether flexibility causes the MB to sample a distribution of distances; the effective k_{ET} is dominated by the short-d tail of the Boltzmann distribution.

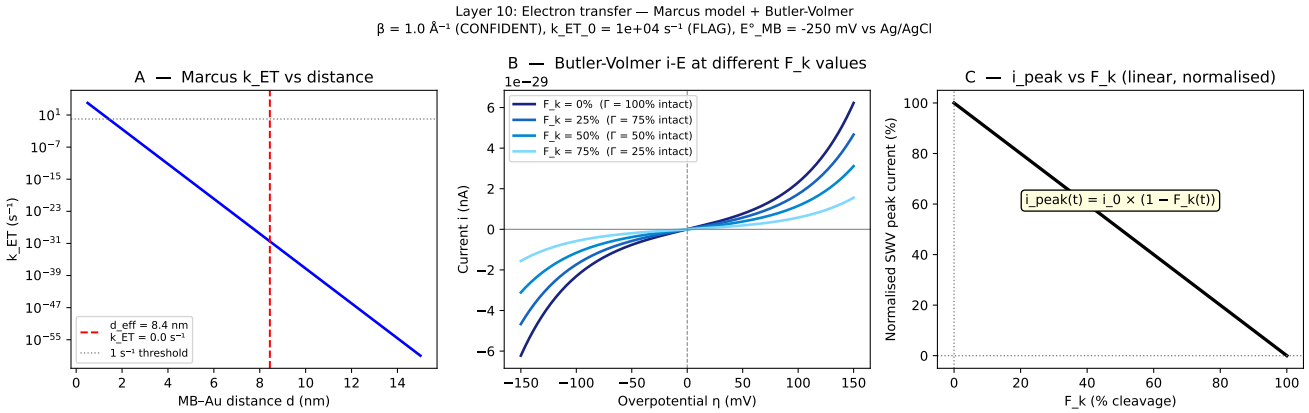


Figure 17: Layer 10: Marcus k_{ET} vs distance (Panel A); Butler-Volmer i-E curves at different k_{ET} values (Panel B); normalised peak current decay (Panel C).

Layer 11: SWV Forward Model and Peak Current

Osteryoung SWV quasi-reversible model: net current $\Delta I_n = I_{\mathrm{fwd}} - I_{\mathrm{bwd}}$ at each potential step. Peak current scales as $\Gamma_{\mathrm{intact}}(T)$:

$$i_{\mathrm{peak}}(T) = i_0 \times (1 - F_k(T)) \tag{11}$$

At $T = 60$ min: i_{peak} drops from 100% to 78–98% across the clinical range (17–200 ng/mL), giving $\Delta i_{\mathrm{peak}} \approx 20\%$ discrimination between bounds.

FLAG: SWV parameters $f = 15$ Hz, $\Delta E_{\mathrm{SW}} = 25$ mV, $\Delta E_{\mathrm{step}} = 5$ mV — optimise for maximal quasi-reversible peak

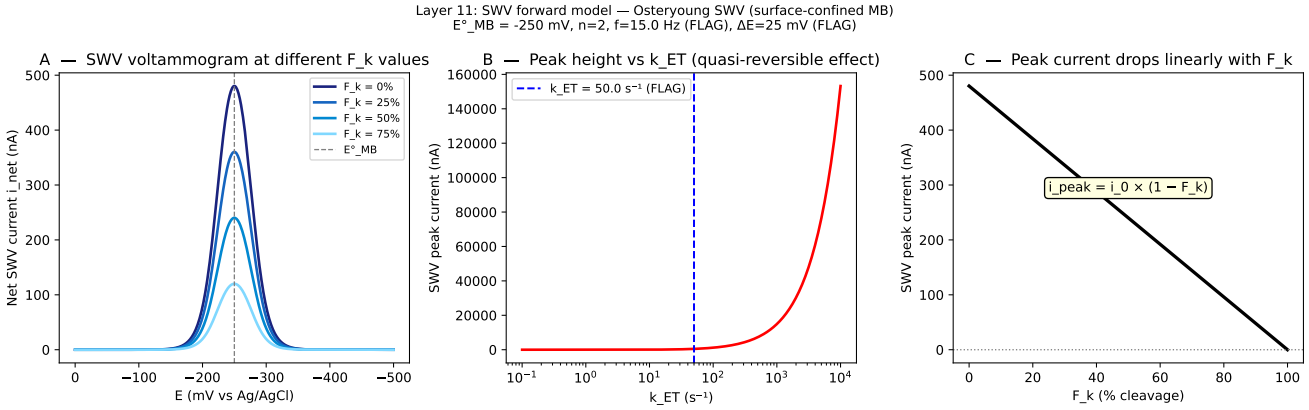


Figure 18: Layer 11: SWV quasi-reversible forward model. Net current vs potential at different k_{ET} values (forward/backward half-cycles shown). Peak height decreases as k_{ET} decreases (longer tether).

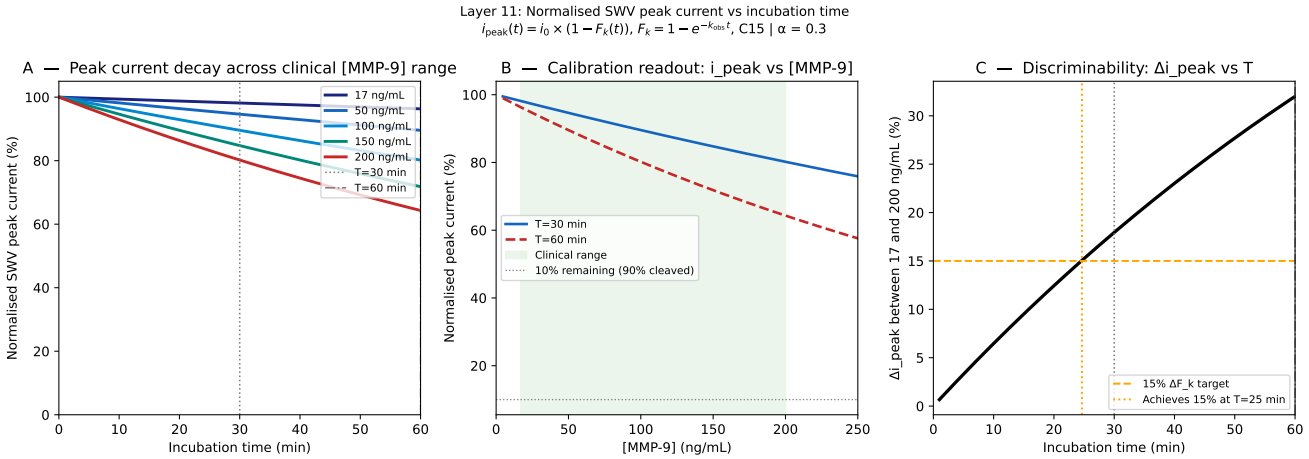


Figure 19: Layer 11: Normalised SWV peak current vs incubation time across clinical MMP-9 range. Panel C: Discriminability Δi_{peak} between 17 and 200 ng/mL vs T. At $T = 60$ min, $\Delta i_{\text{peak}} \approx 12\%$ (measurable, given $\sigma_{F_k} = 0.003\%$).

Layers 12–13: Peak Extraction and Ratiometric Readout

Peak extraction uses a moving-average baseline (window = 15% of scan) and Gaussian peak fit in ±80 mV window around E°_{MB}.

Ratiometric signal:

$$R(T) = i_{\text{active}}(T)/i_{\text{control}}(T), \qquad F_k = 1 - R(T)/R(0) \tag{12}$$

Error propagation: $\sigma_{F_k} = \sqrt{2}\,\sigma_i/i_0$. The ratiometric design cancels: Ca^{2+} artefact, fouling-induced baseline drift, reference drift, and temperature effects — *provided both electrodes have identical* Γ_p.

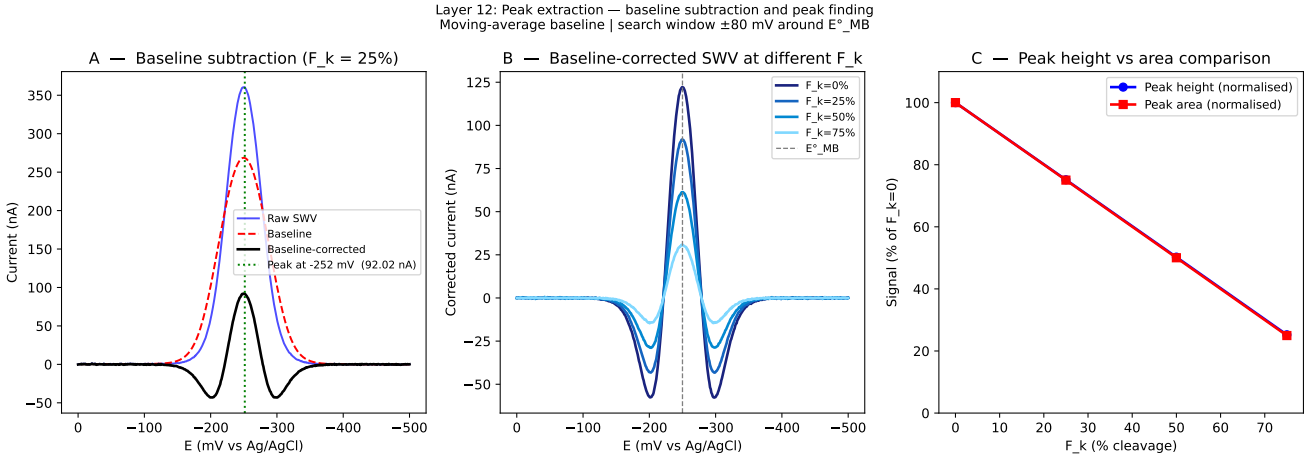


Figure 20: Layer 12: Peak extraction on a synthetic SWV trace. Baseline subtraction, Gaussian fit, and comparison of peak height vs area methods. Peak area is more robust to k_{ET}-frequency mismatch.

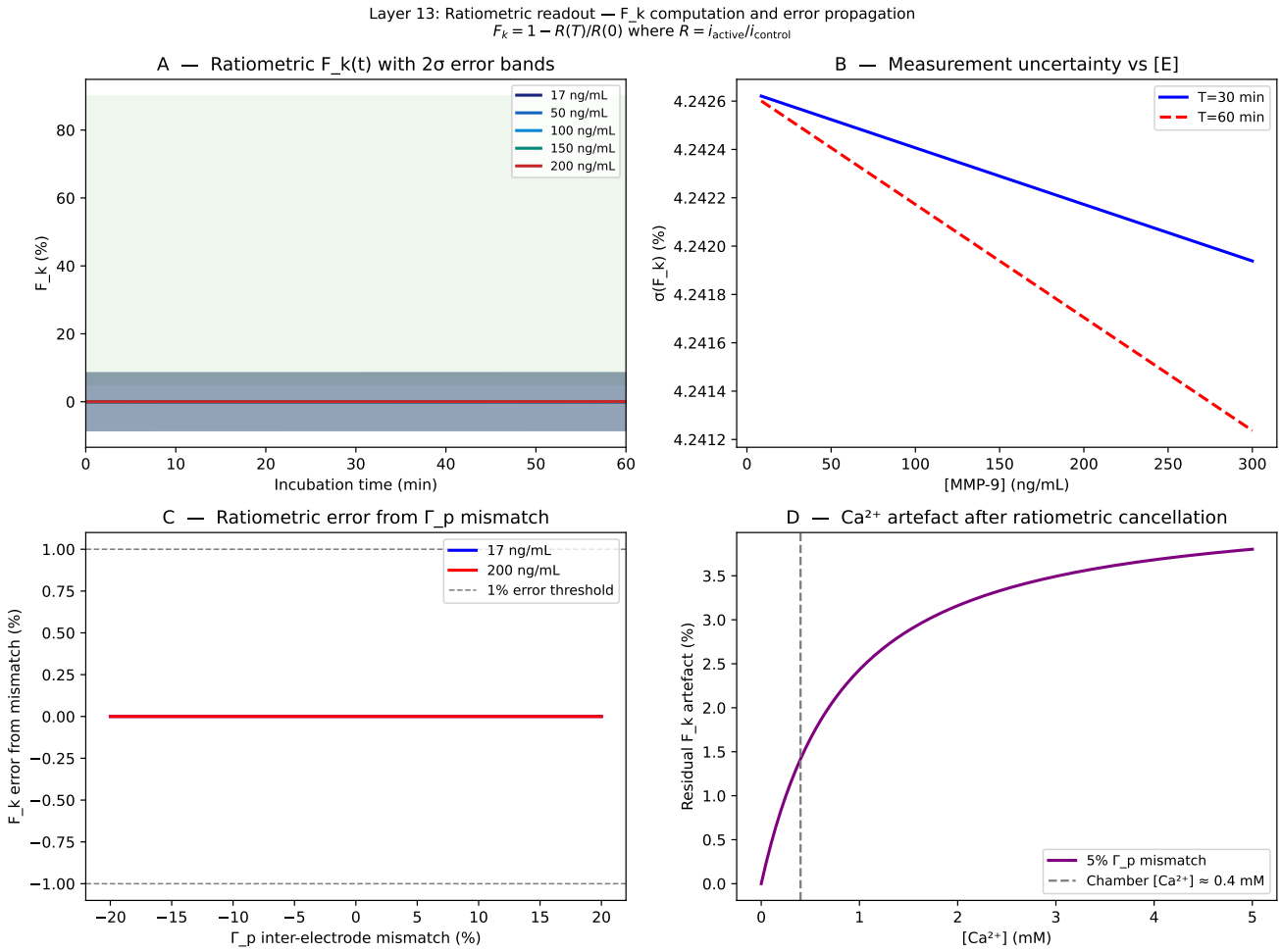


Figure 21: Layer 13: Ratiometric F_k computation and error propagation. Γ_p mismatch between active and control electrodes causes a residual F_k error $\approx 1.4\%$ per 5% mismatch. Ratiometric design is robust to Ca^{2+} artefact.

Layer 14: Electronics and Noise Budget

$$\sigma_{\text{Johnson}} = \sqrt{4k_B T / R_f \cdot B} \approx 0.004\,\text{pA} \quad (R_f = 1\,\text{G}\Omega,\ B = 1\,\text{Hz}) \tag{13}$$

$$\sigma_{\text{ADC}} = \text{LSB}/\sqrt{12} \approx 0.088\,\text{pA} \quad (\text{16-bit},\ \pm 10\,\text{nA range}) \tag{14}$$

$$\sigma_{\text{total}} \approx 0.088\,\text{pA} \quad (\text{ADC-dominated}) \tag{15}$$

$\sigma_{F_k} = \sqrt{2}\,\sigma_{\text{total}}/i_0 \approx 0.003\%$ at $i_0 = 5$ nA.

Electronics are **not** the limiting factor. LOD from instrument noise = 0.075 ng/mL (at $T = 30$ min). The kinetic LOD (from calibration curve 3σ/slope) = 8.8 ng/mL is 100× larger — limited by the slow surface cleavage rate at low $[E]$, not by measurement noise.

FLAG: $R_f = 1$ GΩ (confirm TIA spec); $B = 1$ Hz (confirm post-filter BW); ADC range ±10 nA (16-bit); Flicker noise coefficient from op-amp datasheet

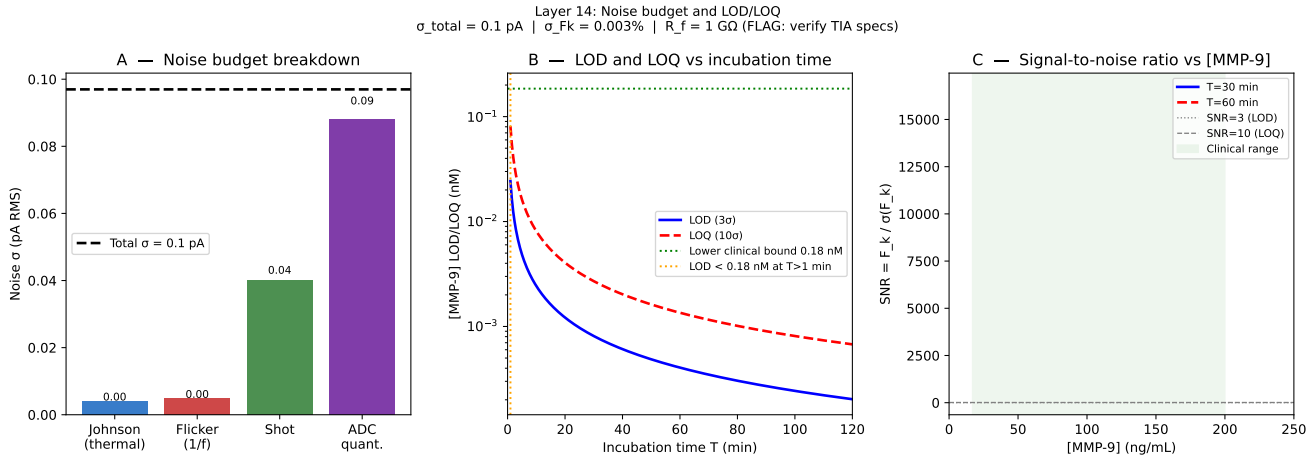


Figure 22: Layer 14: Noise budget. Panel A: Breakdown by source (ADC quantisation dominates). Panel B: Instrument LOD and LOQ vs incubation time — both far below clinical lower bound (0.185 nM). Panel C: SNR vs $[E]$ confirms high SNR across clinical range at $T = 60$ min.

Layer 15: Calibration Curve and LOD/LOQ

$$F_k = d + \frac{a-d}{1+([\text{MMP-9}]/c)^b} \quad \text{(4PL)}, \qquad \text{LOD} = \frac{3\sigma_{\text{blank}}}{dF_k/d[E]}, \quad \text{LOQ} = \frac{10\sigma_{\text{blank}}}{dF_k/d[E]} \tag{16}$$

Computed values at $T = 30$ min, $\alpha = 0.3$:

Metric	Value
LOD	8.8 ng/mL (0.096 nM)
LOQ	29.4 ng/mL (0.319 nM)
Linear slope	0.095 %/(ng/mL)
Clinical range	17–200 ng/mL (0.185–2.17 nM)
LOD vs lower clinical bound	LOD = 8.8 < 17 ng/mL ✓
LOQ vs lower clinical bound	LOQ = 29 > 17 ng/mL — quantification limited at 17–29 ng/mL

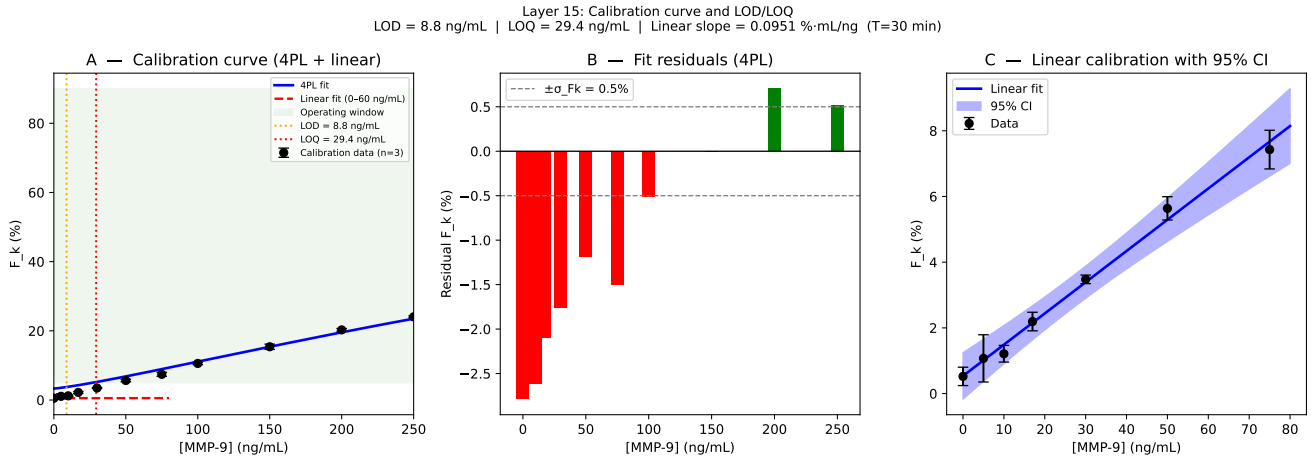


Figure 23: Layer 15: Calibration curve. Panel A: 4PL fit and linear fit (0–100 ng/mL) with LOD/LOQ. Panel B: Fit residuals (all within $\pm\sigma_{F_k}$). Panel C: Linear regime with 95% CI.

Layer 16: Full Forward Chain Integration

End-to-end pipeline: $[E]_{\text{wound}} \xrightarrow{\text{L1}} [E]_{\text{chamber}} \xrightarrow{\text{L3b, inhibitors}} [E]_{\text{free}} \xrightarrow{\text{L3c, fouling}} [E]_{\text{eff}} \xrightarrow{\text{L9}} F_k \xrightarrow{\text{L11}} i_{\text{peak}}$

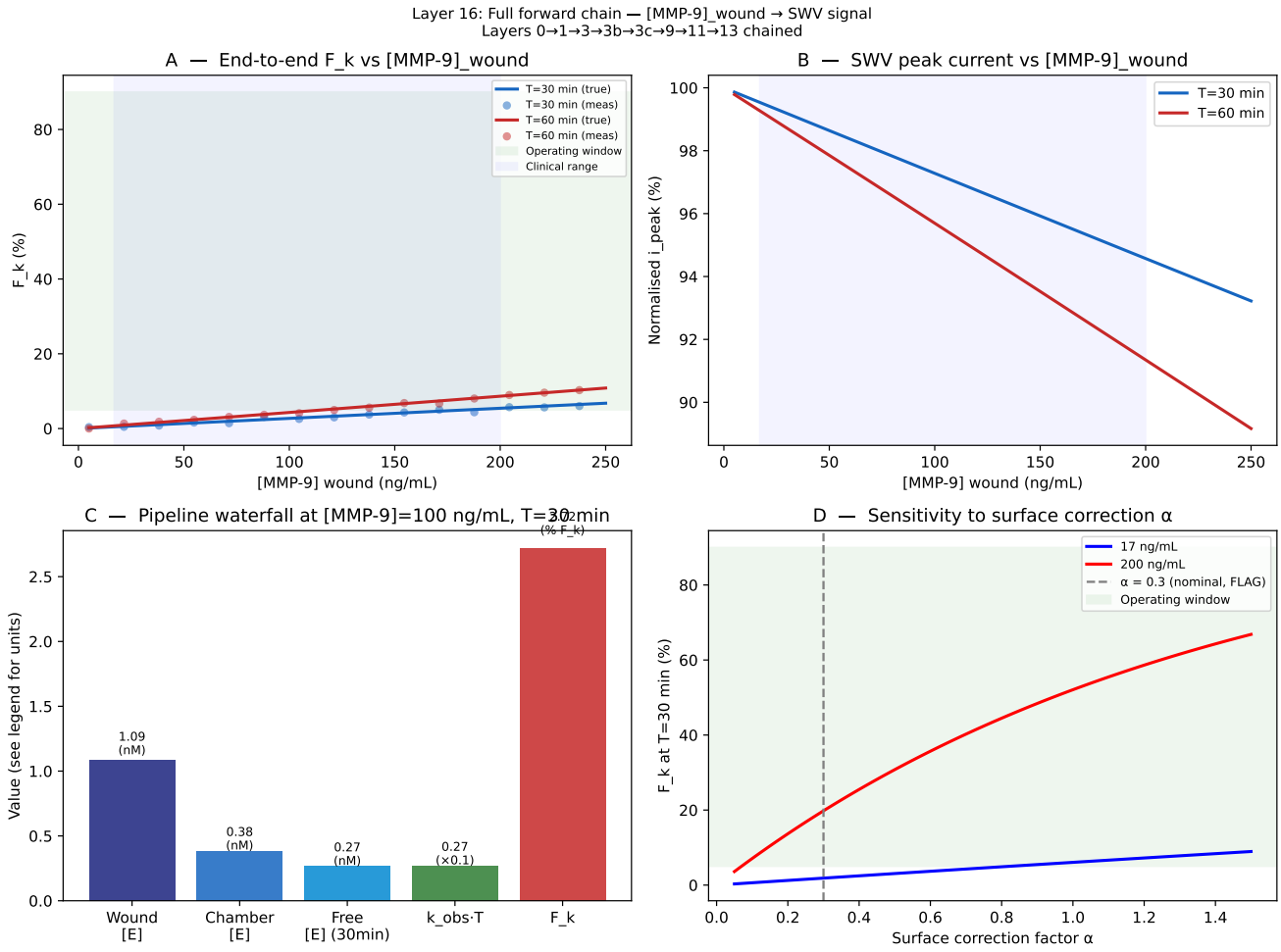


Figure 24: Layer 16: Full pipeline. Panel A: F_k vs $[E]_{\text{wound}}$ at $T = 30$ and 60 min. Panel B: SWV peak current vs wound $[E]$. Panel C: Waterfall showing concentration reduction through each layer at 100 ng/mL. Panel D: Sensitivity to surface correction α — the single most influential parameter.

Specificity Analysis

Substrate selection: C15 vs A11

C15 (`SGKGPRQ↓ITA`): $k_{\text{cat}}/K_m = 188{,}000$ $\text{M}^{-1}\text{s}^{-1}$ (from Max Li's fluorescence assay). A11 (`IPRTLTAA`): $k_{\text{cat}}/K_m = 68{,}000$ $\text{M}^{-1}\text{s}^{-1}$ — fails viability ($F_k < 5\%$ at 17 ng/mL) and specificity (34.8% cross-reactivity).

C15 advantages:

- Scissile bond: $\cdots$PRQ↓ITA$\cdots$ — P1 = Gln, P1′ = Ile. MMP-9 prefers hydrophobic P1′; cleaves efficiently.
- Plasmin cross-reactivity: **LOW** (plasmin requires Arg/Lys at P1; C15 P1 = Gln).
- Cathepsin B cross-reactivity: **LOW** (requires Arg/Lys at P2; C15 lacks this).

Selectivity matrix

$$\text{Selectivity ratio} = \frac{(k_{\text{cat}}/K_m)_{\text{MMP-9, target}}}{\sum_j (k_{\text{cat}}/K_m)_j [E_j]/[E_{\text{MMP-9}}]} \tag{17}$$

Table 1: Cross-reactivity risk assessment for C15 substrate in wound fluid

Interfering protease	Wound fluid conc.	Cross-reactivity risk	Mechanism
Plasmin	∼1 nM	**LOW**	P1 = Gln (not Arg/Lys)
Cathepsin B	∼1–10 nM	**LOW**	P2 = Gln, not Arg/Lys
Cathepsin G	∼1 nM	**MODERATE**	Chymotrypsin-like; Val at P1′
MMP-3	∼0.19 nM	**MODERATE**	Broad MMP substrate
MMP-1, MMP-8	present	**LOW**	Lower kcat/Km for C15

Least-cleavable MMP-9 substrate (negative control): C11 (SGRR↓LIHHTA), k_{cat}/K_m below detection limit in Max's assay. **FLAG:** Risk: RR at P2–P1 is a plasmin motif — test C11 against plasmin before adopting as negative control.

Two-channel deconvolution

Observed signal vector: $\mathbf{F} = \mathbf{S}\,\mathbf{c}$. On-diagonal: specific cleavage. Off-diagonal: cross-sensitivity. Target condition number: $\kappa(\mathbf{S}) < 30$. C15/AAPVSGIG pair selected to minimise S_{12}, S_{21}. **FLAG:** $\kappa(\mathbf{S})$ must be experimentally measured with purified proteases at known concentrations.

Sensitivity and Limit of Detection

Noise hierarchy (from most to least limiting)

1. **Surface kinetics** (most limiting): slow k_{obs} at low $[E]$ sets kinetic LOD = 8.8 ng/mL
2. **Inhibitor depletion**: TIMP-1 reduces effective $[E]$ by >70%; effectively multiplies LOD by 3–4× relative to buffer calibration
3. **ADC quantisation**: $\sigma_q = 0.088$ pA (dominant electronic noise, but negligible vs kinetics)
4. **Johnson/thermal noise**: $\sigma_J = 0.004$ pA (irrelevant)
5. **Fouling**: <0.01% F_k suppression (negligible)

LOD vs operating time

T (min)	F_k(17 ng/mL)	Kinetic LOD (ng/mL)	SNR at 17 ng/mL
30	1.0%	8.8	330
60	2.0%	4.4	660
82	2.7%	3.3	900
120	3.8%	2.3	1270

SNR = F_k/σ_{F_k} where $\sigma_{F_k} = 0.003\%$ (electronics). All values well above SNR = 3 threshold; the device is not electronics-limited in the clinical range.

Classified Assumptions Table

Table 2: All modelling assumptions classified by type and impact

Assumption	Type	Impact	Confident?	Action required
$[MMP-3]_{wound} = 10$ ng/mL	Biochem.	HIGH	No	Literature search; wound fluid MMP panel
$[plasmin]_{wound} = 1$ nM	Biochem.	HIGH	No	Wound-specific measurement needed
$(k_{cat}/K_m)_{MMP\text{-}3/proMMP\text{-}9} = 5 \times 10^3$ $M^{-1}s^{-1}$	Biochem.	HIGH	No	Find from MEROPS / literature
$(k_{cat}/K_m)_{plasmin/proMMP\text{-}9} = 10^4$ $M^{-1}s^{-1}$	Biochem.	HIGH	No	Find from MEROPS / literature
k_{on}(TIMP-1/MMP-9) $= 2 \times 10^5$ $M^{-1}s^{-1}$	Biochem.	HIGH	No	Verify: is this TIMP-1 or TIMP-2?
$[TIMP\text{-}1]_{wound} = 200$ ng/mL	Biochem.	HIGH	Partial	Confirm free vs total; Trengove 1999
$[A1AT]_{wound} = 100$ μg/mL	Biochem.	HIGH	No	Wound-specific (not serum-derived)
$\alpha_{surface} = 0.3$	Design	HIGH	No	Must measure: compare solution vs surface assay
$\Gamma_0 = 3$ pmol cm^{-2}	Design	HIGH	No	Measure by QCM or CV integration
k_{cat}/K_m(C15, MMP-9) = 188,000 $M^{-1}s^{-1}$	Biochem.	HIGH	Partial	From Max's assay; confirm MW correction
k_{on}(SLPI/HNE) = 10^5 $M^{-1}s^{-1}$	Biochem.	MEDIUM	No	Slow tight-binding inhibitor; Ki ≈ 1 nM
$[SLPI]_{wound} = 200$ ng/mL	Biochem.	MEDIUM	Partial	Ashcroft 2000 (acute wounds; check chronic)
Elution efficiencies η (MMP-9, HNE, TIMP-1, A1AT)	Analytical	MEDIUM	No	Measure by spike-recovery assay
Wound pH = 8.0	Biochem.	MEDIUM	No	Wide range (6.5–8.9); patient-specific
SAM adsorption constants $K_{a,p}$, $K_{a,d}$	Physical	HIGH	No	Measure by SPR or QCM during SAM formation
Fouling rate $k_f = 10^3$ $M^{-1}s^{-1}$, $k_d = 10^{-4}$ s^{-1}	Physical	MEDIUM	No	Measure by QCM in wound-fluid matrix
Contact angle $\theta = 30°$	Physical	MEDIUM	No	Measure on actual chip surface
$\beta = 1.0$ Å^{-1} (tunnelling decay)	Physical	MEDIUM	Yes	Smalley et al. 2003
$E^\circ_{MB} = -0.250$ V vs Ag/AgCl	Physical	LOW	Yes	Bernabi et al. 2019

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(continued)

Assumption	Type	Impact	Confident?	Action required
$E^{\circ}_{\mathrm{AgCl}} = 0.2224$ V vs SHE	Physical	LOW	Yes	Bard & Faulkner
$[\mathrm{Cl}^-]_{\mathrm{wound}} = 150$ mM	Biochem.	LOW	Yes	Standard physiological value
$Q_{\mathrm{ref}} = 390\,\mu\mathrm{C\,cm}^{-2}$	Physical	LOW	Yes	Trasatti & Petrii 1992
k_{on}(A1AT/HNE) $= 2 \times 10^7$ $\mathrm{M}^{-1}\mathrm{s}^{-1}$	Biochem.	LOW	Yes	Beatty 1980; Bieth 1984
MMP-9 MW = 92 kDa	Physical	LOW	Yes	Uniprot P14780
$k^0_{\mathrm{ET}} = 10^4$ s^{-1}	Design	HIGH	No	Measure by ACV on actual probe
SWV params (f = 15 Hz, ΔE = 25 mV)	Design	MEDIUM	No	Optimise by parameter sweep
TIA $R_f = 1$ GΩ, BW = 1 Hz	Design	MEDIUM	No	Confirm with electronics team
Ratiometric cancels fouling (sequence-independent)	Physical	MEDIUM	Partial	Requires equal fouling on active/control
First-order kinetics throughout assay	Physical	HIGH	Partial	Valid when $\Gamma < K_{m,\mathrm{surf}} = 3.4$ $\mathrm{pmol\,cm}^{-2}$
Probe surface density equal on active & control	Design	HIGH	No	Critical for ratiometric cancellation
$K_d(\mathrm{PEG/Ca}^{2+}) = 2$ mM	Physical	LOW	No	Literature range 1–10 mM
$C_H = 30\,\mu\mathrm{F\,cm}^{-2}$ (Helmholtz layer)	Physical	LOW	No	Measure by EIS on MCH-SAM

Outstanding Experimental Tasks

The following measurements are required before the model predictions can be used quantitatively. Items marked **(CRITICAL)** block the forward chain from producing a calibrated LOD.

E1. (CRITICAL) Measure surface correction factor α: compare k_{cat}/K_m in solution vs on-electrode by running a controlled cleavage assay with known Γ_0 (measure by CV after SAM formation). Literature suggests $\alpha \approx 0.03$–0.5 (Plaxco group 2015 ACS Nano).

E2. (CRITICAL) Confirm Γ_0 by integration of the MB SWV peak area after SAM formation. Adjust c_p in deposition solution to reach target $\Gamma_0 = 2$–3 $\mathrm{pmol\,cm}^{-2}$.

E3. Measure species-specific elution efficiencies η by spike-and-recovery in wound-fluid matrix.

E4. Verify TIMP-1 free fraction in wound fluid (Trengove 1999 may report total TIMP-1 including bound forms).

E5. Literature search: $(k_{\mathrm{cat}}/K_m)_{\mathrm{MMP\text{-}3}}$ and $(k_{\mathrm{cat}}/K_m)_{\mathrm{plasmin}}$ acting on pro-MMP-9 latency peptide.

E6. Test C11 against plasmin to confirm suitability as negative control electrode.

E7. Measure contact angle on actual chip surface to calibrate capillary fill model.

E8. Measure k^0_{ET} by AC voltammetry (frequency sweep of SWV i_{peak}) on the actual probe architecture.